Bloc A

#pe #apunts #english #PARCIAL1

1.1. Basic Definitions & Set Operations

- **Deterministic vs. Random Phenomena**: Deterministic phenomena always lead to the exact same results under the same initial conditions, whereas random phenomena involve uncertainty in the future outcome.
- **Sample Space (Ω) & Events**: The sample space Ω is the set of all possible outcomes. An **event** is any subset of Ω.
- **Partition**: A collection of mutually disjoint events $A_i \neq \emptyset$ whose union equals the full sample space Ω ($A_i \cap A_j = \emptyset$ for $i \neq j$ and $\bigcup A_i = \Omega$).
- **Set Operations**: Events follow standard set operations including **Union** ($A \cup B$), **Intersection** ($A \cap B$), **Complement** ($\neg A$), and **Difference** ($A - B$).
- **Representations**: Probability problems are visually represented using **Venn diagrams** or **Probability trees**.

1.2. Probability Axioms & Properties

- **Axioms**: A probability function assigns a value between 0 and 1 to each event such that:
  1. $0 \leq P(A) \leq 1$
  2. $P(\Omega) = 1$
  3. $P(\bigcup A_i) = \sum P(A_i)$ for disjoint events
- **Deducted Rules**:
  - **Complement Rule**: $P(\neg A) = 1 - P(A)$
  - **Empty Set**: $P(\emptyset) = 0$
  - **Inclusion-Exclusion Principle**: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
  - **Equiprobability**: $P(A) = \frac{\text{favourable cases}}{\text{total cases}}$ applies **only** when all outcomes have equal probability.

1.3. Conditional Probability & Multiplicative Rule

- **Conditional Probability**: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ (for $P(B) > 0$) measures the probability of A occurring given that B has already occurred. Conditioning on B effectively reduces the sample space to B.
- **Multiplicative Rule**: $P(A \cap B) = P(A) \cdot P(B|A)$. In a probability tree, the joint probability along any path is the product of the probabilities along its branches6.

1.4. Law of Total Probability & Bayes' Theorem

- **Law of Total Probability (LTP)**: Given a partition $A_1, A_2, \ldots, A_$ of Ω, the probability of an event B_k is calculated by summing across all partition paths:

$$P(B_k) = \sum_j P(B_k|A_j) \cdot P(A_j)$$

- **Bayes' Formula**: Reverses conditional probabilities:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- **Bayes' Theorem**: Combining Bayes' formula with the LTP gives:

$$P(A_i|B_k) = \frac{P(B_k|A_i) \cdot P(A_i)}{\sum_j P(B_k|A_j) \cdot P(A_j)}$$

1.5. Event Independence

- **Definition**: A and B are **independent** if and only if:

$$P(A \cap B) = P(A) \cdot P(B)$$

- **Equivalences**: If A and B are independent, knowing B occurred does not change the likelihood of A, meaning $P(A|B) = P(A)$ and $P(B|A) = P(B|\neg A) = P(B)$

2.1. Definitions: Discrete (DRVs) vs. Continuous (CRVs)

- **Random Variable (RV)**: A function $X : \Omega \to \mathbb{R}$ that assigns a real numerical value to each outcome in the sample space Ω.
- **Discrete Random Variable (DRV)**: Takes a countable set of values (e.g., counts, integers).
  - **Probability Function (PMF)**: $p_X(k) = P(X = k)$, where $\sum_k p_X(k) = 1$.
  - **Cumulative Distribution Function (CDF)**: $F_X(k) = P(X \leq k) = \sum_{j \leq k} p_X(j)$.
- **Continuous Random Variable (CRV)**: Takes values from an uncountably infinite set (usually real numbers or intervals).
  - **Probability Density Function (PDF)**: A non-negative function $f_X(x) \geq 0$ where the total area under f_X equals 1 ($\int_{-\infty}^{+\infty} f_X(x)dx = 1$). **Crucial note**: Point probabilities for CRVs are zero ($P(X = k) = 0$), so $f_X(k)$ is a height/density, not a point probability.
  - **Cumulative Distribution Function (CDF)**: $F_X(k) = P(X \leq k) = \int_{-\infty}^{k} f_X(x)dx$, with $f_X(x) = \frac{dF_X(x)}{dx}$.

2.2. Interval Probabilities & Quantiles

- **Interval Probabilities**:
  - **CRVs**: Strict vs. non-strict inequalities do not alter probabilities because single points have measure zero: $P(a \leq X \leq b) = P(a < X < b) = F_X(b) - F_X(a) = \int_a^b f_X(x)dx$.
  - **DRVs**: Boundaries matter! $P(a < X \leq b) = F_X(b) - F_X(a)$, but $P(a \leq X \leq b) = F_X(b) - F_X(a-1)$.
- **Quantiles (x_α)**: The value x_α such that the cumulative probability up to x_α equals α ($F_X(x_\alpha) = \alpha$). Finding a quantile is the inverse problem of finding cumulative probabilities ($x_\alpha = F_X^{-1}(\alpha)$).

2.3. Indicators: Expectation & Variance

- **Expected Value (Mean μ_X or $E(X)$)**:
  - **DRV**: $E(X) = \sum_k k \cdot p_X(k)$.
  - **CRV**: $E(X) = \int_{-\infty}^{+\infty} x \cdot f_X(x)dx$.
- **Variance (σ_X^2 or $V(X)$) & Standard Deviation (σ_X)**:
  - Measures dispersion about the mean.
  - **Key Computational Formula**: $V(X) = E(X^2) - [E(X)]^2$.
- **Linear Transformations & Properties**:
  - **Shift & Scale**: $E(a + bX) = a + bE(X)$ and $V(a + bX) = b^2V(X)$ (adding a constant a shifts the mean but does not alter variance).
  - **Sum of Expectations**: $E(X + Y) = E(X) + E(Y)$ (always holds, regardless of independence).
  - **Sum of Variances**: $V(X \pm Y) = V(X) + V(Y)$ **only if** X and Y are independent (notice it is always a "+" sign on the right side).

2.4. Bivariate Random Variables & Association

- **Joint & Conditional Functions**:
  - **Joint PMF / PDF**: Describes the joint behavior $p_{X,Y}(x, y) = P(X = x \cap Y = y)$ for DRVs or $f_{X,Y}(x, y)$ for CRVs.
  - **Conditional Functions**: $p_{X|Y=y}(x) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$ or $f_{X|Y=y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$.
- **Independence of RVs**:
  - X and Y are independent if and only if $p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y)$ (or $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$).
  - When independent, $E(X \cdot Y) = E(X) \cdot E(Y)$ and $\mathrm{cov}(X, Y) = 0$.
- **Covariance ($\mathrm{cov}(X, Y)$) & Pearson Correlation ($\rho_{X,Y}$)**:
  - **General Expectation Formula**: $E(X \cdot Y) = E(X) \cdot E(Y) + \mathrm{cov}(X, Y)$
  - **General Variance Formula**: $V(X \pm Y) = V(X) + V(Y) \pm 2\mathrm{cov}(X, Y)$
  - **Covariance Properties**: $\mathrm{cov}(aX, bY) = a \cdot b \cdot \mathrm{cov}(X, Y)$ and $\mathrm{cov}(X, X) = V(X)$

- **Correlation Coefficient**: Standardized indicator of linear relationship bounded between -1 and 1:

$$\rho_{X,Y} = \frac{v(X,Y)}{\sigma_X \cdot \sigma_Y}$$

- **Linearity Rules**: If $|\rho_{X,Y}| = 1$, the relationship is perfectly linear ($Y = a + bX$). If X, Y are independent, then $\rho_{X,Y} = 0$; however, $\rho_{X,Y} = 0$ does **not** imply independence (it only implies the absence of a *linear* relationship).

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Set Operations

```r
A <- c(1, 2, 3, 4)
B <- c(3, 4, 5, 6)
Omega <- 1:10 # Define the full sample space

# Union (A U B)
union(A, B)

# Intersection (A ^ B)
intersect(A, B)

# Difference (A - B)
setdiff(A, B)

# Complement (¬A)
setdiff(Omega, A)
```

Law of Total Prob. & Bayes' Theorem

```r
# --- STEP 1: Define Priors and Conditional Likelihoods ---
# P(A_i): Prior probabilities of partition causes (A1, A2, A3)
P_A <- c(0.50, 0.30, 0.20)

# P(B | A_i): Likelihood of observing event B given each cause
# P(B | A_i) =  P(B ^ A) / P(A)
P_B_given_A <- c(0.02, 0.05, 0.10)

# Check Axiom: sum of priors must equal 1
sum(P_A) == 1

# --- STEP 2: Calculate Joint Probabilities P(B ∩ A_i) ---
# Multiplicative rule along each tree path: P(B ∩ A_i) = P(A_i) * P(B |
A_i)
P_joint <- P_A * P_B_given_A
```

```r
# --- STEP 3: Law of Total Probability (LTP) ---
# P(B) = Sum of all joint path probabilities
P_B <- sum(P_joint) cat("Total Probability P(B) =", P_B, "\n")

# --- STEP 4: Bayes' Theorem P(A_i | B) ---
# P(A_i | B) = P(B ∩ A_i) / P(B)
P_A_given_B <- P_joint / P_B

# --- STEP 5: Display as a Summary Table ---
bayes_table <- data.frame(
    Cause = paste0("A", 1:length(P_A)),
    Prior_P_A = P_A,
    Likelihood_P_B_given_A = P_B_given_A,
    Joint_Path = P_joint,
    Posterior_Bayes = P_A_given_B
)

print(bayes_table)
```

DRVs

Defining a DRV & Checking Axioms

```r
# Values of X and their probabilities
x <- c(0, 1, 2, 3, 4)
p_x <- c(0.10, 0.25, 0.35, 0.20, 0.10)

# Check axiom: sum of probabilities must equal 1
sum(p_x) == 1
```

CDF, Fx

```r
# Cumulative probabilities P(X >= k)
F_x <- cumsum(p_x)

# Evaluate P(a < X <= b) = F_X(b) - F_X(a)
# Example: P(1 < X <= 3) = F_X(3) - F_X(1)
prob_interval <- F_x[which(x == 3)] - F_x[which(x == 1)]
```

Mean $E(X)$ and Variance $V(X)$

```r
# Expected Value E(X) = sum(x * p(x))
E_X <- sum(x * p_x)

# Mean of X^2: E(X^2) = sum(x^2 * p(x))
E_X2 <- sum(x^2 * p_x)
```

```r
# Variance V(X) = E(X^2) - (E(X))^2

V_X <- E_X2 - (E_X)^2

# Standard Deviation sigma_X = sqrt(V(X))
sd_X <- sqrt(V_X)

cat("E(X) =", E_X, "| V(X) =", V_X, "| SD =", sd_X, "\n")
```

CRVs

Checking Validity of Density Function $f_X(x)$

```r
# Define the PDF f(x)
f_x <- function(x) {
    # Example: f(x) = 2*x for 0 <= x <= 1, 0 otherwise
    ifelse(x >= 0 & x <= 1, 2 * x, 0)
}

# Total area must equal 1
total_area <- integrate(f_x, lower = 0, upper = 1)$value
cat("Total area =", total_area, "\n")
```